



HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 1

CANDIDATE
NAME

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CT GROUP

12S7_____

CENTRE
NUMBER

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INDEX
NUMBER

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BIOLOGY

8875/01

Paper 1 Multiple Choice

27 September 2013

Additional Materials: Optical Mark Sheet

1 hour

INSTRUCTIONS TO CANDIDATES

1. Write your **name**, **CT group**, **Centre number** and **index number** in the spaces provided at the top of this cover page.
2. Fill in your particulars on the Optical Mark Sheet. Write your **NRIC number** and shade accordingly.
3. There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.
4. At the end of the paper, you are to submit **only** the Optical Mark Sheet.

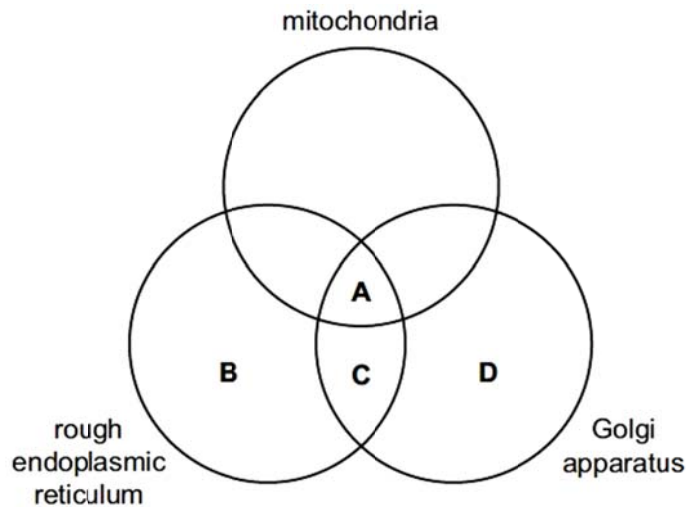
INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

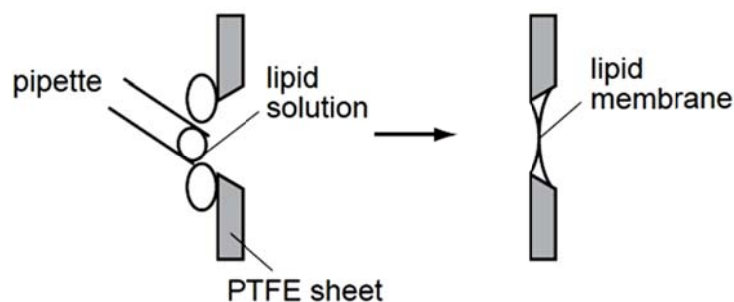
Any rough working should be done in this booklet.

Calculators may be used.

- 1 Which organelle(s) is / are required for the formation of the hydrolytic enzymes found in lysosomes?



- 2 Lipid membranes can be formed in the laboratory by painting phospholipids over a support (PTFE sheet) with a hole in it.



Such a lipid membrane is impermeable to water-soluble materials including charged ions such as Na^+ or K^+ . In one experiment with Na^+ ions, no current flowed across the membrane until a substance called gramicidin was added.

Which statement is consistent with this information and your knowledge of membrane structure?

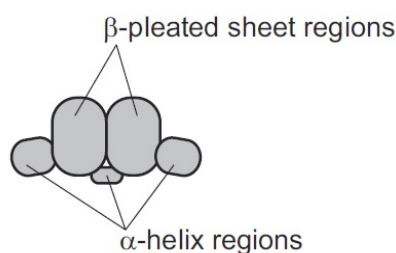
Gramicidin becomes incorporated into the membrane and is

- A a carbohydrate molecule found only on the outside of the membrane.
- B a non-polar lipid which passes all the way through the membrane.
- C a protein molecule with both hydrophilic and hydrophobic regions.
- D a protein molecule which has only hydrophobic regions.

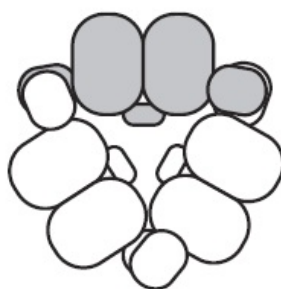
- 3 Approximately half of the total protein in a pea seed consists of the storage protein vicilin.

Each molecule of vicilin is made up of three identical polypeptides.

Each polypeptide is made up of two β -pleated sheet regions with linking α -helix regions, folded into the shape shown to the right.



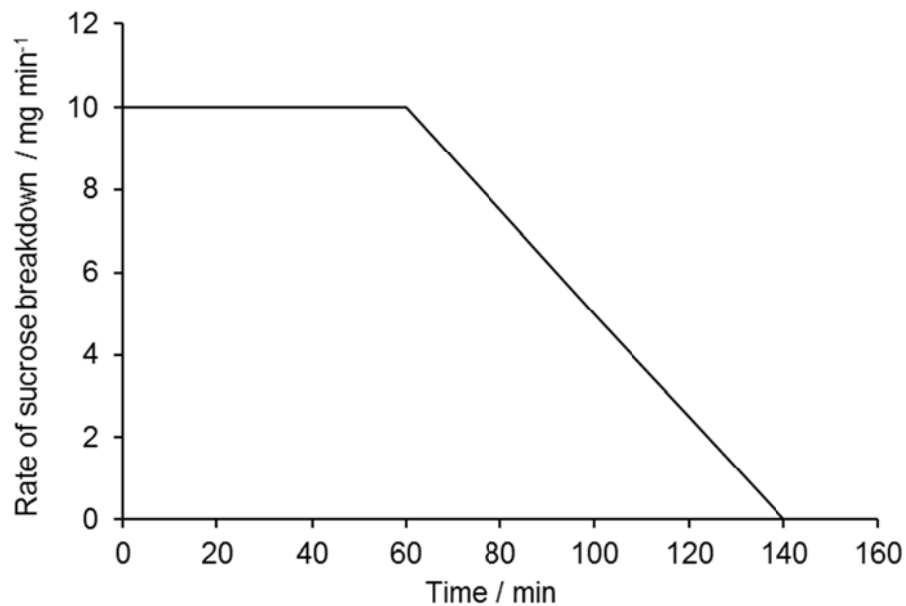
This allows the three polypeptides to pack together into a compact, flat storage molecule, as shown below.



Which row correctly describes the structure of vicilin?

	primary structure	secondary structure	tertiary structure	quaternary structure
A	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	association of three polypeptides	folding of each polypeptide
B	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	folding of each polypeptide	association of three polypeptides
C	association of three polypeptides	amino acid sequence of one polypeptide	α -helix and β -pleated sheet regions of each polypeptide	folding of each polypeptide
D	association of three polypeptides	amino acid sequence of one polypeptide	folding of each polypeptide	α -helix and β -pleated sheet regions of each polypeptide

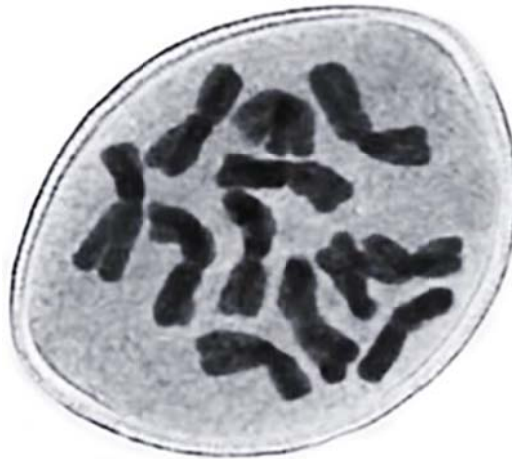
- 4 The graph shows the results of an investigation using the enzyme invertase that breaks down sucrose into glucose and fructose. 1 g of sucrose was dissolved in 100 cm³ water and 2 cm³ of a 1 % invertase solution was added.



Which conclusion can be drawn from this information?

- A Between 0 and 60 min, the concentration of the substrate remains constant.
- B After 60 min, the concentration of enzymes becomes the limiting factor.
- C At 140 min, some of the enzyme molecules are denatured.
- D Between 60 and 140 min, the concentration of the substrate is the limiting factor.

- 5 The diagram shows a stage of mitosis in a cell. The amount of DNA present is 12 picograms (pg).



Which row correctly identifies the number of DNA molecules and amount of DNA in each nucleus at telophase of mitosis?

	number of DNA molecules	amount of DNA / pg
A	12	6
B	24	12
C	12	12
D	24	6

- 6 Nocodazole is a chemical used in the study of mitosis. It causes all mitotic cells to be arrested at metaphase.

Which statement(s) correctly identify how nocodazole might work?

- 1 Inhibits chromatin condensing in the nucleus
- 2 Prevents replication of the centrioles
- 3 Stops sister chromatids migrating to opposite poles

- A** 3 only **B** 1 and 2 only **C** 1 and 3 only **D** All of the above

- 7 Exposure of DNA to ultraviolet (UV) light can result in the covalent linking of adjacent thymine bases on the same strand of DNA, forming a T-dimer.

Individuals with the genetic disease xeroderma pigmentosum have a greatly increased risk of developing cancer in skin that has been exposed to UV light. This is because the normal repair of T-dimers by an excision endonuclease enzyme (excinuclease) does not occur.

Which individual is most likely to develop skin cancer?

individual	presence of adjacent thymine bases in DNA	mutation in gene for excinuclease
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

- 8 A double-stranded DNA molecule is 10 kb long.

Which row correctly describes the structure of this DNA molecule?

	number of nucleotides	number of complete turns	length of DNA molecule / μm
A	10000	1000	3.4
B	20000	1000	0.34
C	20000	1000	3.4
D	20000	10000	3.4

- 9 An alien organism was found and investigated. Like eukaryotes, it has a linear double-stranded DNA genome. When DNA replication was examined, it revealed that although the process is semi-conservative, no Okazaki fragments were observed in all the multiple replication forks. In addition, the end-replication problem was not observed.

Which statement correctly explains this phenomenon?

- A Alien DNA is antiparallel.
 - B Alien DNA replication only starts at the 3' end of each template strand.
 - C Alien DNA polymerase synthesizes DNA in both 5' to 3' and 3' to 5' directions.
 - D Alien DNA ligase is not involved in the DNA replication process.
- 10 You have three different double-stranded DNA molecules, each of which you incubate (separately) in solution that is heated from 25 °C to 100 °C to denature the double-stranded molecules.

Molecule 1 is 1000 bp long and has a G-C content of 70 %.

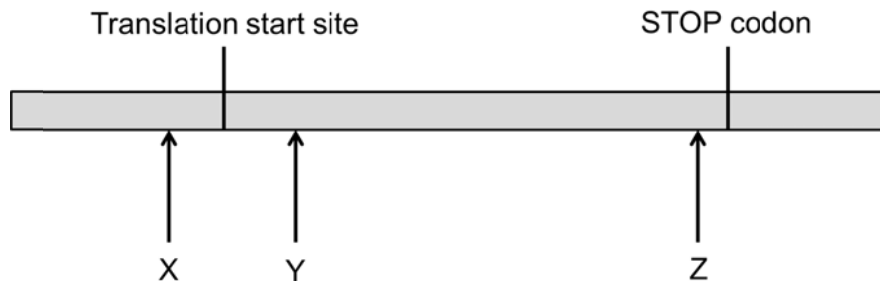
Molecule 2 is 1000 bp long and has a G-C content of 50 %.

Molecule 3 is 2000 bp long and has a G-C content of 40 %.

Which statement is likely to be the correct prediction of the denaturation of these molecules?

- A Molecule 2 will denature at the highest temperature.
 - B Molecule 3 will denature at the highest temperature.
 - C Molecules 1 and 2 will denature at a higher temperature than Molecule 3.
 - D All three molecules will denature at the same temperature.
- 11 Which statements about the genetic code are correct?
- 1 The genetic code has redundancy and is degenerate.
 - 2 There is only one codon for the amino acid methionine.
 - 3 Codons act as 'stop' and 'start' signals during transcription and translation.
 - 4 Prokaryotes generally use the same genetic code as eukaryotes.
 - 5 mRNA codons have the same nucleotide sequence as DNA triplet codes.
- A 1, 2 and 3 only B 1, 2 and 4 only C 1, 3 and 5 only D 2, 4 and 5 only

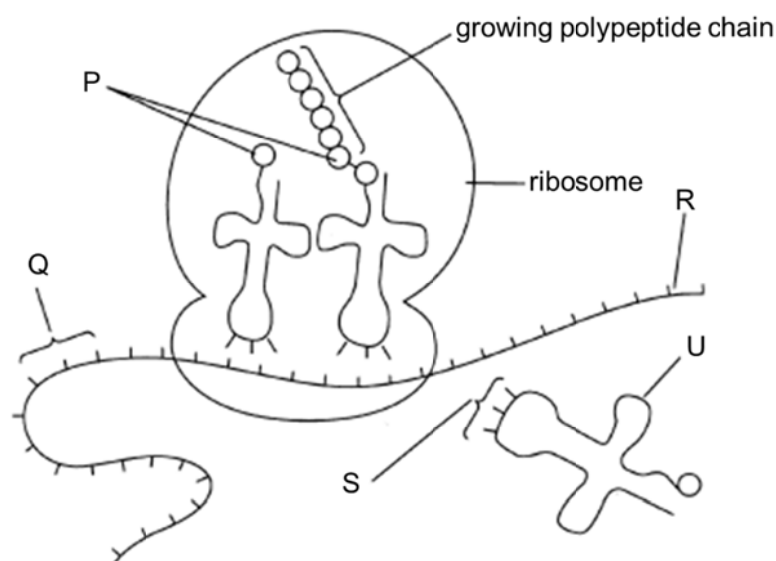
- 12** During the process of transcription, errors sometimes occur such that certain nucleotides are repeated. These errors are often referred to as duplications. Duplications of nucleotide bases within genes vary in severity. Some result in non-functional proteins and others may have little or no effect on function. The diagram shows a strand of mRNA produced from a particular gene.



Which duplication event is most likely to lead to the synthesis of a non-functional protein?

- A** One base pair is duplicated at Z.
- B** Three base pairs are duplicated at X.
- C** One base pair is duplicated at Y.
- D** Three base pairs are duplicated at Y.

- 13** The diagram illustrates the synthesis of the protein insulin in genetically engineered yeast cells.



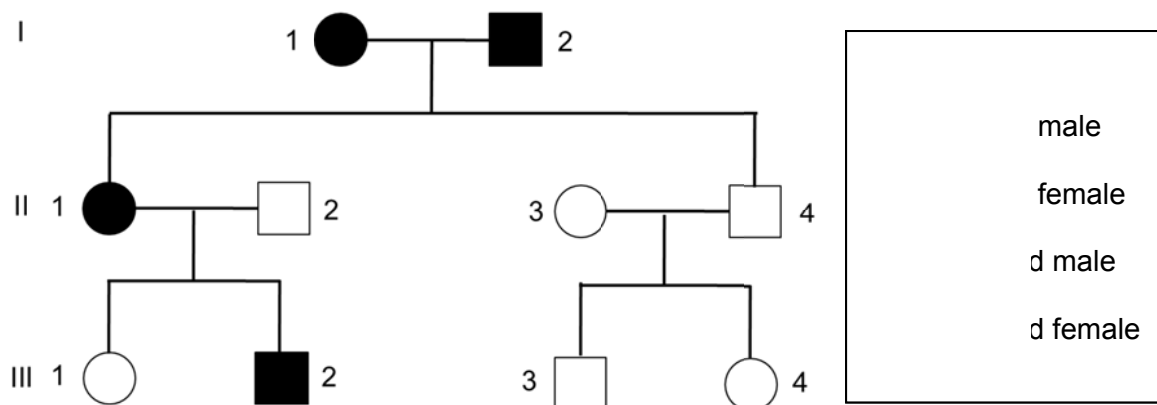
Which statement is correct?

- A** P cannot be used for synthesis of ribosome.
 - B** Protein synthesis always proceed in the direction of R to Q.
 - C** Q can bind to many S.
 - D** U will form peptide bonds with ribosomes during synthesis of the polypeptide chain.
- 14** A population of cats contains individuals with shortened tails known as Manx cats and individuals with normal tails known as non-Manx cats. Mating of two non-Manx cats always produced non-Manx cats. Mating of two Manx cats always produced a mixture of Manx cats and non-Manx cats, where the number of Manx cats is usually twice that of non-Manx cats.

Which hypothesis is consistent with these observations?

- A** The genes for shortened tails and normal tails are linked on the same chromosome.
- B** The genes for shortened tails and normal tails are sex-linked.
- C** The allele for shortened tails is dominant over the allele for normal tails, but the homozygous dominant genotype is lethal.
- D** The allele for shortened tails is dominant over the allele for normal tails, but the homozygous recessive genotype is lethal.

- 15 The pedigree shows inheritance of a dominant trait.



Which identifies the correct probabilities of the trait appearing in the offspring for the two marriages?

	probability of trait inheritance	
	marriage 1: III - 1 x III - 3	marriage 2: III - 2 x III - 4
A	0 %	50 % offspring
B	50 % offspring	0 %
C	0 %	50 % female offspring
D	50 % male offspring	50 % female offspring

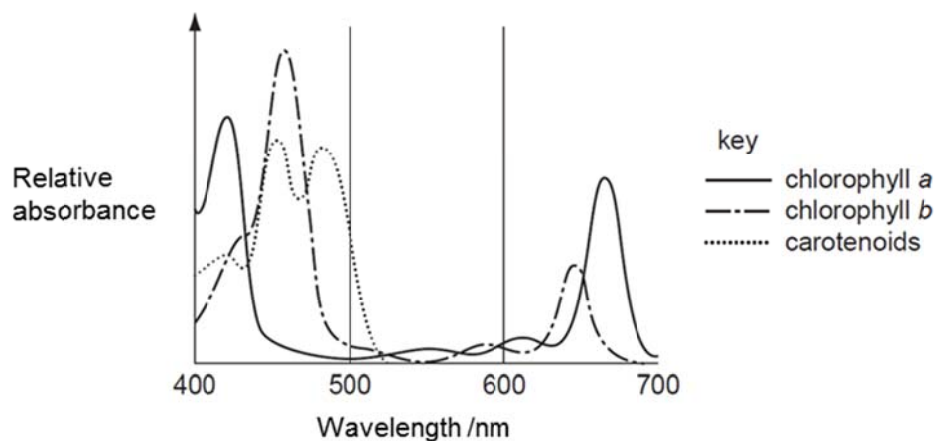
- 16 In a breed of rabbits, multiple alleles with the following dominance relationships control coat colouration: C (agouti) $> c^{ch}$ (chinchilla) $> c^h$ (Himalayan) $> c$ (albino)

An experimental cross between agouti and Himalayan produced 50 % agouti and 50 % Himalayan progeny.

Which cross would have the lowest probability of producing this result?

- A** $Cc^h \times c^h c^h$ **B** $Cc \times c^h c$ **C** $Cc^h \times c^h c$ **D** $Cc \times c^h c^h$

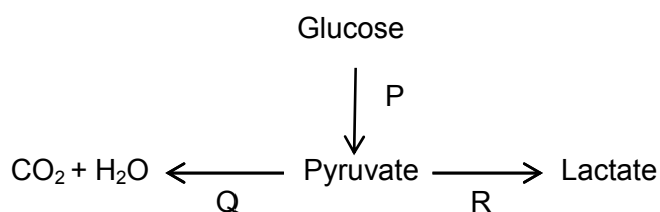
- 17 The graph shows the absorption spectra of some pigments found in chloroplasts.



Which statement is **not** correct?

- A Having several pigments, rather than one, increases the efficiency of photosynthesis.
- B Most leaves are green as chlorophyll absorbs light in the blue and red regions of the spectrum.
- C Photosynthesis will be fastest in light at the red end of the spectrum, as red light has higher energy than blue light.
- D Prior to leaf fall, chlorophyll is broken down, leaving carotenoids which makes leaves look yellow or red.

- 18 The diagram shows the relationship between various metabolic processes.



Which statements are correct?

- 1 Acetyl coenzyme A is required for step Q.
- 2 Substrate level phosphorylation occurs in both steps P and Q.
- 3 There is a net increase in ATP during steps P, Q and R.
- 4 Oxidised form of NAD is used in steps P and R.

- A 1 and 2 only B 1 and 3 only C 2 and 4 only D 1, 2 and 4 only

- 19** Mammalian liver cells were homogenised and the resulting homogenate centrifuged. Portions containing only mitochondria and cytosol (residual cytoplasm) were each isolated. Samples of each portion, and of the complete homogenate, were incubated in four ways:

- 1 with glucose;
- 2 with pyruvate;
- 3 with glucose plus cyanide; and
- 4 with pyruvate plus cyanide.

Cyanide inhibits carriers in the electron transport chain. After incubation, the presence or absence of carbon dioxide and lactate in each sample was determined. The results are summarised in the table below.

	samples of homogenate					
	complete		mitochondria only		cytosol	
	carbon dioxide	lactate	carbon dioxide	lactate	carbon dioxide	lactate
1 glucose	✓	✓	×	×	×	✓
2 pyruvate	✓	✓	✓	×	×	✓
3 glucose + cyanide	×	✓	×	×	×	✓
4 pyruvate + cyanide	×	✓	×	×	×	✓

× = absent, ✓ = present

Which statement can be concluded from the table?

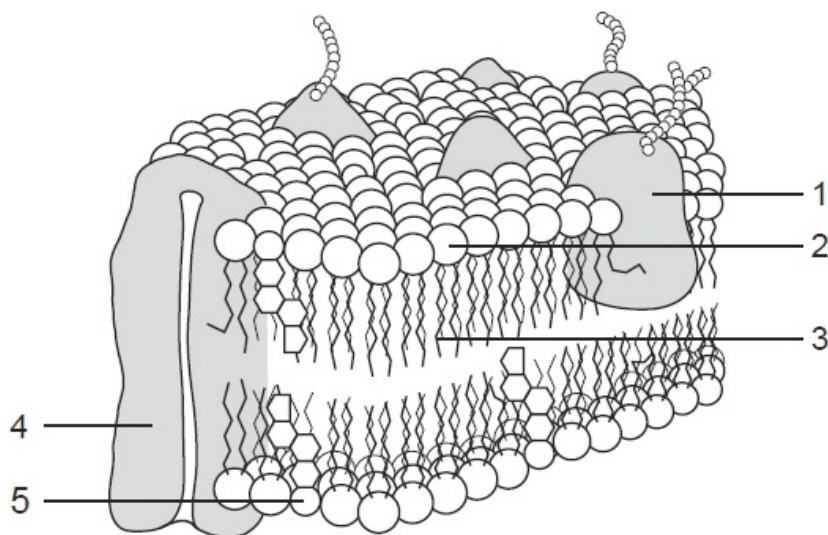
- A** Carbon dioxide was not formed when mitochondria was incubated with glucose as there was no oxygen present.
- B** Both aerobic and anaerobic respiration were occurring in 3.
- C** Lactate formation in mitochondria was inhibited by the presence of cyanide.
- D** The mode of action of cyanide would be similar to that of a cell experiencing anaerobic respiration.

- 20 The table shows the results of food tests carried out to identify two types of biomolecules.

test-tube	reagent added to test-tube		
	sodium hydroxide + copper sulphate	Benedict's solution	iodine in potassium iodide
X	purple	brick red	brown
Y	blue	blue	blue-black
Z	purple	blue	blue-black

Which conclusion is consistent with the results?

- A Only albumen and sucrose had been placed in tube X.
 B Only maltose and starch had been placed in tube Z.
 C Only starch and sucrose had been placed in tube Y.
 D Only starch and sucrose had been placed in tube Z.
- 21 The diagram shows part of the cell surface membrane.



Which components help to maintain the fluidity of the membrane?

- A 1 and 3 only
 B 1 and 4 only
 C 2 and 4 only
 D 3 and 5 only

- 22** A population's gene pool was found to have remained unaltered for many generations.

Which condition must have existed in the population?

- A** mating had always been random
- B** genetic drift had often occurred
- C** immigration and emigration were common
- D** inbreeding had been commonly practiced

- 23** A strain of bacterium which causes sore throats was found to be susceptible to a particular antibiotic. An individual with a sore throat recovered after taking a course of the antibiotic. Several weeks later when the same individual again developed a sore throat caused by the same strain of bacterium, a second course of the same antibiotic was ineffective.

Which statement is the most probable explanation for the ineffectiveness of the second course of antibiotic?

- A** Resistance to antibiotic developed in the non-resistant bacteria after exposure to the first course of antibiotic.
- B** Antibiotic caused a mutation in the non-resistant bacteria which made the bacteria resistant to the antibiotic.
- C** Antibiotic acted as a selecting agent for resistance in the bacterial population.
- D** No resistant bacteria existed in the population before penicillin was used.

- 24** Before the settlement of California in the 1800s, the elk population was very large. By about 1900 there were only a few dozen elk left. Owing to protection, there are now about 3000 elk living in a small number of isolated herds. Unfortunately, some of the elk in all the herds have difficulty grazing due to a shortened lower jaw.

Which statements best explain this?

- 1 The early settlers only hunted elk that could graze.
- 2 There was a mutation affecting jaw size.
- 3 There is random mating within each herd.
- 4 The current elk population demonstrates genetic drift.
- 5 There was directional selection favouring short jaws.

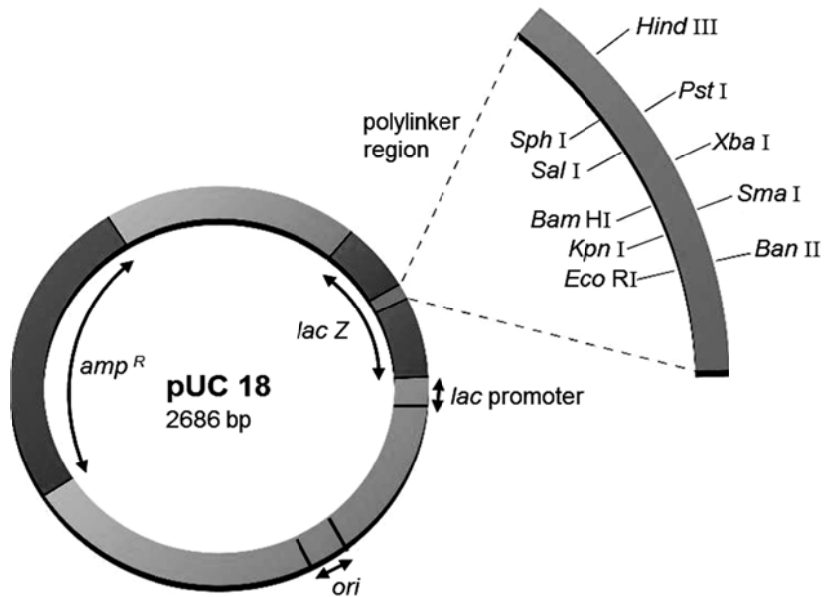
A 1, 2 and 4 only **B** 1, 3 and 5 only **C** 2, 3 and 4 only **D** 2, 3 and 5 only

- 25** *Hind* III is a restriction endonuclease commonly used to cut human DNA into pieces before inserting it into a plasmid.

Which is most likely to be the recognition sequence for this enzyme?

- A** AAGGAA
B AAGAAG
C AAGTTC
D AAGCTT

26 The diagram shows a vector, pUC18, used in genetic engineering.

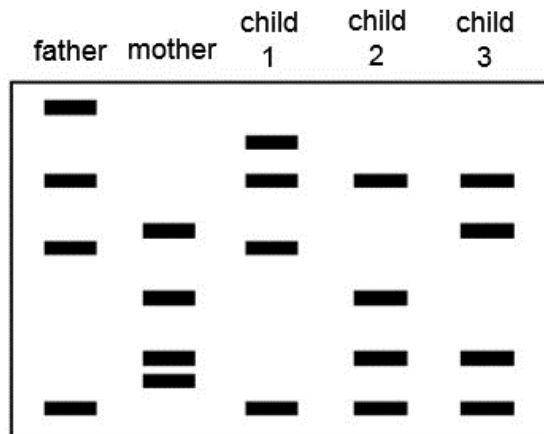


Which statements are true for the use of pUC18 in genetic engineering?

- 1 Gene of interest is inserted into the *lac* promoter.
- 2 Gene of interest is inserted into the polylinker.
- 3 Bacteria colony with successfully inserted gene of interest will appear white in colour.
- 4 Bacteria colony with this vector will grow in agar plate with ampicillin.

A 1 and 3 only **B** 2 and 4 only **C** 1, 2 and 4 only **D** 2, 3 and 4 only

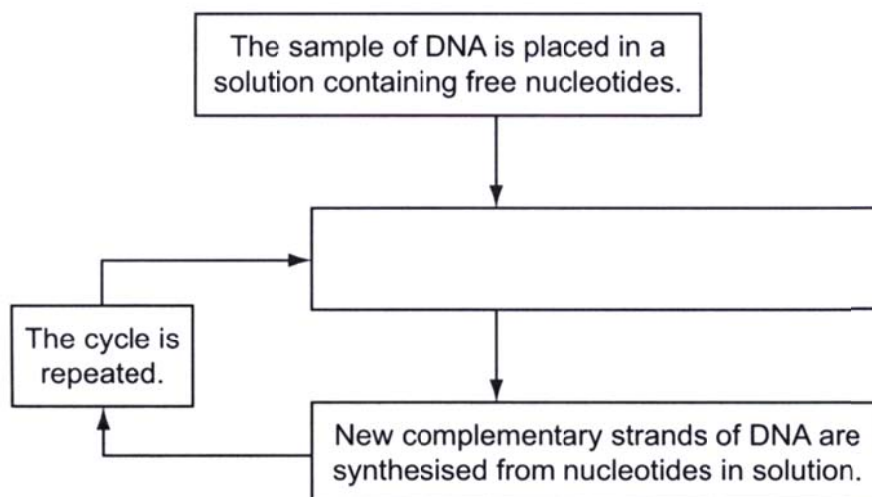
- 27 DNA fingerprinting is often used to confirm the identity of an individual. The diagram shows a small section of the gel electrophoresis results from a DNA fingerprinting analysis of a family.



Which of the children is least likely to be the offspring of both parents?

- A child 1
- B child 2
- C child 3
- D all of the children are definitely offspring of both parents

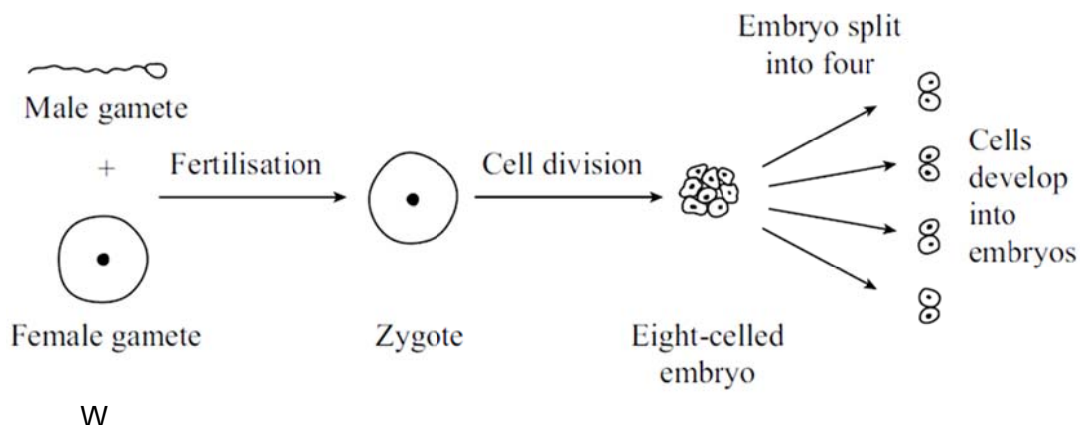
28 The polymerase chain reaction is summarised in the flowchart below.



Which statement completes the flow chart?

- A Complementary strands of DNA are separated.
- B Free nucleotides join on the end of DNA strands.
- C Small sections of DNA are formed.
- D Strands of DNA bind to RNA primers.

- 29 The diagram shows various stages in the development of embryos.



Which statements are correct?

- 1 Cell W is multipotent.
- 2 Cell X is totipotent.
- 3 Cell Y can undergo self-renewal.
- 4 Cell Z is pluripotent.

- A** 1 and 2 only **B** 3 and 4 only **C** 2, 3 and 4 only **D** All of the above

- 30 What are the arguments against the use of genetically modified organisms (GMOs)?

- 1 Insufficient testing of genetically modified crop for their side effects
- 2 Unforeseen long-term effects of genetic manipulation
- 3 Accidental genetic recombination in humans as a result of consuming food derived from GMOs
- 4 Control of food supply by a small number of companies that have access to genetic engineering technology

- A** 1 and 2 only **B** 2 and 3 only **C** 1, 2 and 3 only **D** All of the above

--- END OF PAPER ---